NADUN DE SILVA

Lead Software Engineer

Auckland, New Zealand
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Summary

Lead Software Engineer with 8.5+ years architecting and building large-scale distributed systems and cloud-native SaaS backends, while setting technical direction for multi-cloud platforms on Kubernetes, Amazon Web Services (AWS), and Google Cloud Platform (GCP). Proven track record of leading and mentoring engineering teams of up to 7, driving end-to-end system design decisions, and delivering resilient systems into production.

Experience

Lead Software Engineer

McCrae Tech | April 2025 to Present | Auckland, New Zealand

- Owned the technical direction and cloud-native architecture of Indexity's automated customer environment provisioning across 8 AWS accounts and 2 GCP projects, guiding 2 engineers from design through delivery.
- Architected a test environment provisioning and result-aggregation framework on Amazon EKS using Python and Terraform, reducing setup time by 90% and establishing it as the engineering standard across the Orchestral platform.
- Engineered an automated failure-detection and alerting system for the Indexity data-planes with Datadog and Terraform, surfacing customer-impacting issues within 15 minutes and eliminating customer-reported incidents.
- Devised Indexity's database and service upgrade strategy on AWS, constraining the self-managed OrientDB upgrade to under 10 minutes of downtime with a rollback plan that de-risked the cutover.
- Directed the zero-downtime handover of Indexity's AWS accounts from Orion Health to McCrae Tech, with an account-sequencing plan that kept all customer environments continuously available.

Senior Software Engineer

Orion Health | November 2022 to April 2025 | Auckland, New Zealand

- Designed and built Indexity's AWS cloud architecture and disaster-recovery strategy that reduced downtime risk by at least 40%, delivering a resilient, highly available SaaS platform to customers.
- Mentored 2 junior engineers building features for the Indexity SaaS platform, setting its technical direction and driving design and code reviews that upheld platform quality.
- Hardened the Indexity platform against security risks through comprehensive threat modeling and privacy assessments, achieving zero major vulnerabilities across all customer-initiated penetration tests.
- Enhanced phone number search by designing and implementing Apache Lucene-based indexing techniques, improving patient search result precision by 30% across all customers.
- Constructed an end-to-end deployment framework for Indexity on AWS and a Terraform-provisioned self-hosted GitLab runner on Google Cloud, cutting costs by over 50% and reducing each delivery cycle by a day.

Associate Technical Lead

WSO2 | June 2021 to November 2022 | Colombo, Sri Lanka

- Steered a product team of 7 engineers on Choreo, WSO2's SaaS platform, delivering 95% of team targets and sprint milestones on time through clear technical prioritization and a collaborative engineering culture.
- Led a Senior Software Engineer through the design and delivery of the resource-scheduling feature for Choreo's browser-based cloud IDE within 1.5 months using Kubernetes and Go.
- Secured the Sustained Outstanding Contribution Award for three consecutive years, an honor reserved for the top 5% of staff, recognizing exceptional technical leadership and innovation.
- Scaled concurrent user capacity on the Choreo platform by introducing a Redis cache with randomized expiry times, slashing MSSQL database utilization by 60%.
- Eliminated bottlenecks in Choreo's cloud IDE, reducing startup times by 80% and improving the experience for users building on the platform through Kubernetes resource optimization.

Senior Software Engineer

WSO2 July 2019 to June 2021 Colombo, Sri Lanka

- Spearheaded the design and implementation of Choreo observability features within 3 months, leading a team of 2 engineers to deliver the platform's core monitoring and tracing capabilities.
- Minimized production bugs across Choreo's Observability product domain by establishing code-review, testing, and deployment standards for a team of 6 engineers, influencing engineering practices across peer teams.
- Established Choreo observability storage and data archival strategies using a Data Lake, reducing company infrastructure costs by 90% and enabling long-term data retention for Artificial Intelligence (AI) use cases.
- Extended WSO2's open-source Ballerina compiler to generate instrumentation mapping Choreo observability data to source code for applications deployed on the platform, improving debugging for Choreo users within 1 month.

Software Engineer

WSO2 January 2018 to July 2019 Colombo, Sri Lanka

- Championed the development of developer tooling for users deploying applications with Cellery, an open-source Cell-based application deployment framework, using VS Code Language Server Extensions and D3.js visualizations.
- Delivered the foundational Cellery observability features in 2 months, enabling visibility into microservice composites using Kubernetes, Istio, OpenTracing, and Envoy.
- Built Cellery Hub, a registry for Cellery composite images, using a Docker Registry as the backend and implemented a browser-based OpenID Connect (OIDC) authentication flow for the Cellery CLI and portal within 1 month.
- Implemented the observability capabilities of the WSO2 Serverless Platform, enabling end-users to monitor their serverless functions using Prometheus and Jaeger on Kubernetes and OpenWhisk.

Skills

- **Programming languages**: Java, Go, Scala, Python, JavaScript, TypeScript
- **Data Storage**: Time-series Databases (InfluxDB, Azure Data Explorer), Graph Databases (OrientDB), Data Lakes (Azure Data Lake), Relational Databases (MySQL, Microsoft SQL Server), Redis
- **API Protocols**: REST, GraphQL, gRPC
- **Cloud Platforms**: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
- **Containerization**: Kubernetes (K8s), Azure Kubernetes Service (AKS), Amazon Elastic Container Service (ECS), Amazon Elastic Kubernetes Service (EKS), Docker, Kustomize, Helm
- **Frameworks and tools**: React, ExpressJS, Kubernetes Operator Framework, Terraform, Ansible
- **Software Development Methodologies**: Agile, DevOps
- **AI Coding Tools**: Claude Code, Cursor, Antigravity
- **Observability Tools and Platforms**: OpenTelemetry, Datadog, Prometheus, Jaeger, CloudWatch
- **CI/CD Tools and Practices**: GitOps, GitHub Actions, GitLab

Education

B.Sc. (Hons.) in Engineering (Computer Science and Engineering)

University of Moratuwa January 2014 to December 2017 Colombo, Sri Lanka

- Attained a GPA of 3.85 out of 4.20, obtaining a First Class.
- Placed on the Dean's List in 6 out of 8 semesters at the University of Moratuwa.
- Named Global Finalist (Galactic Impact) in the NASA Space Apps Challenge 2017.
- Completed Google Summer of Code 2017.

Certifications

Certified Kubernetes Administrator

The Linux Foundation December 2020

Certified Kubernetes Application Developer

The Linux Foundation January 2020